

Open-Data Machine Learning for Streamflow and Hydrological Drought Forecasting in the Peruvian Altiplano

Richar Andre Vilca-Solorzano and Dina Maribel Yana-Yucra

Abstract Hydrological variability and drought threaten water security in the Peruvian Altiplano, where sparse gauges and proprietary tools limit operational forecasting. We release a fully open machine-learning pipeline for short-horizon streamflow forecasts and a simplified standardized streamflow drought indicator. Quality-controlled daily discharge from Andean-GC is paired with Open-Meteo/ERA5-Land precipitation across a 13-station Peru benchmark (gauges with ≥ 5000 QC days), with Altiplano focus on P00000055 (Puente Isla Cabanillas). Persistence and seasonal climatology are compared with Ridge, random forests, gradient boosting, XGBoost and a small multilayer perceptron under chronological splits and walk-forward checks, reporting RMSE, MAE, NSE, Kling–Gupta efficiency (KGE) and skill versus persistence. At a one-day lead, persistence remains strongest (NSE = 0.902); at seven days, Ridge improves on persistence (NSE = 0.550 vs. 0.462; skill = 0.086), and all 13 gauges show positive best-model skill (median = 0.102). The study is a transparent Andean benchmark, not a replacement for national hydrological services.

Key words: streamflow forecasting, hydrological drought, Altiplano, open data, machine learning, Nash–Sutcliffe efficiency, Kling–Gupta efficiency, multi-station benchmark

Richar Andre Vilca-Solorzano (corresponding author)
Professional School of Statistical and Computer Engineering, Faculty of Statistical and Computer Engineering (FINESI), National University of the Altiplano (UNAP), Puno, Peru.
e-mail: \emailAndre{}. ORCID: \orcidAndre{}

Dina Maribel Yana-Yucra
Professional School of Statistical and Computer Engineering, Faculty of Statistical and Computer Engineering (FINESI), National University of the Altiplano (UNAP), Puno, Peru.
e-mail: \emailMaribel{}. ORCID: \orcidMaribel{}

1 Introduction

Communities and irrigated agriculture in the Peruvian Altiplano depend on highly seasonal streamflow into the Lake Titicaca system. Climate variability, glacier retreat in headwaters and limited real-time tooling make even short-range discharge forecasts valuable for water allocation and drought awareness. Large-sample machine-learning (ML) hydrology has shown that data-driven models can compete with process models when trained carefully [9], yet many published pipelines remain hard to reproduce in Andean settings because gauges, licences and code are fragmented.

This paper contributes a laptop-scale, open-data multi-station benchmark with Altiplano focus, including:

1. a documented provenance inventory (Andean-GC discharge [1] and Open-Meteo precipitation from ERA5-Land [7, 8]) for 13 long Peru gauges;
2. strict temporal validation without leakage, plus expanding-window walk-forward on the focus gauge;
3. honest comparison to persistence and climatology using NSE [2], KGE [3] and skill versus persistence;
4. a documented SSI-30 drought indicator (month-wise z -score of the 30-day mean; see limitations relative to Vicente-Serrano et al. [10]);
5. publication-ready figures and Python code in a public GitHub repository (`\githubUrl{}`).

This hydrology paper continues an open Altiplano research line by the same team: a prior multi-station comparative climate-ML benchmark for frost/ $T_{\min}$ [6], the AgroYachay cloud IoT/LLM platform for Andean smallholders [4], and AgroCom-mish, its companion desktop edge-commissioning tool released as open software on Zenodo [5] (SoftwareX manuscript in preparation; not cited here as a published SoftwareX article). Unlike those farm-IoT systems, the present work is open analysis and short-horizon streamflow/drought forecasting rather than sensor provisioning or farm-node telemetry.

2 Related Work

Data-driven streamflow models are now common in mountain basins. Random forests with engineered satellite and ground precipitation features have reached high NSE at short leads in tropical Andean catchments [11]. In Peru, neural networks have been used for monthly forecasts in high-Andean basins such as Crisnejas [12], and k -nearest neighbours have been applied to daily flows of the Ramis river in the Altiplano [13]. Large-sample LSTM hydrology provides methodological context for benchmarking ML against process baselines [9]. Closely related prior work by the present authors compared twelve statistical, ML and deep-learning models for frost and minimum-temperature prediction across thirteen Altiplano stations [6] (open repository: <https://github.com/Andre031222/>

Table 1 Compact comparison of related Andean/Peru ML streamflow studies versus this open multi-station benchmark. Entries summarise scope as reported in the cited sources; they are not a meta-analysis of reimplemented scores.

Study	Domain	Lead / scale	Persistence KGE	/ Open multi-station
Muñoz et al. [11]	Tropical (Ecuador), RF + satellite precip	Andes Hourly, one catchment	NSE focus; not multi-gauge Peru	No
Ramos et al. [12]	Crisnejas, Peru; ANN	Monthly	Basin-specific ANN	No
Quispe et al. [13]	Ramis (Altiplano), k -NN	Daily, Peru; river	one Single-basin forecast	No
Kratzert et al. [9]	Large-sample LSTM hydrology	Catchment CAMELS-style	Strong lines; Andean gauges	base- Large-sample (non-Peru) open
Torres-Cruz et al. [6]	Altiplano, frost/ $T_{\min}$ benchmark	Peru; Daily, 13 stations	Multi-model skill; not stream-flow	Yes (climate, not discharge)
This study	Peru GC, focus	Andean- Altiplano $h = 1, 7$; 13 gauges	Daily Persistence, NSE, KGE, skill	Yes (open)

frost-prediction-altiplano-puno); that IJACSA study is prior Altiplano multi-station comparative climate ML (frost/ $T_{\min}$), whereas here we focus on open Andean-GC discharge, hydrological drought indication and honest persistence/KGE reporting. In parallel, AgroYachay [4] (<https://github.com/Andre031222/agroyachay>; Zenodo doi:10.5281/zenodo.20829992) provides Andean smallholder IoT/LLM decision support in the cloud, and AgroCommish [5] (<https://github.com/Andre031222/agrocommish>; Zenodo doi:10.5281/zenodo.20655609) is the companion desktop tool for ESP32 edge commissioning (software release; SoftwareX in preparation). This paper is the open hydrology analysis/forecasting branch of that lineage, not a farm-IoT systems paper.

Hydrological drought monitoring often relies on standardized indices. Vicente-Serrano et al. proposed robust computation of the Standardized Streamflow Index (SSI) [10]. Our SSI-30 indicator is a simplified month-wise z -score of rolling discharge for the open benchmark, not a full distributional SSI implementation. Reanalysis precipitation is obtained via Open-Meteo from ERA5-Land [7, 8]. Model skill reporting follows NSE [2], KGE [3] and multi-metric guidelines for watershed models [14].

Table 1 situates this work against prior Andean and Peruvian ML streamflow or drought studies already cited above. The distinctive angle here is an open multi-station Peru benchmark with explicit persistence and KGE honesty at multiple leads, rather than single-basin peak NSE claims alone.

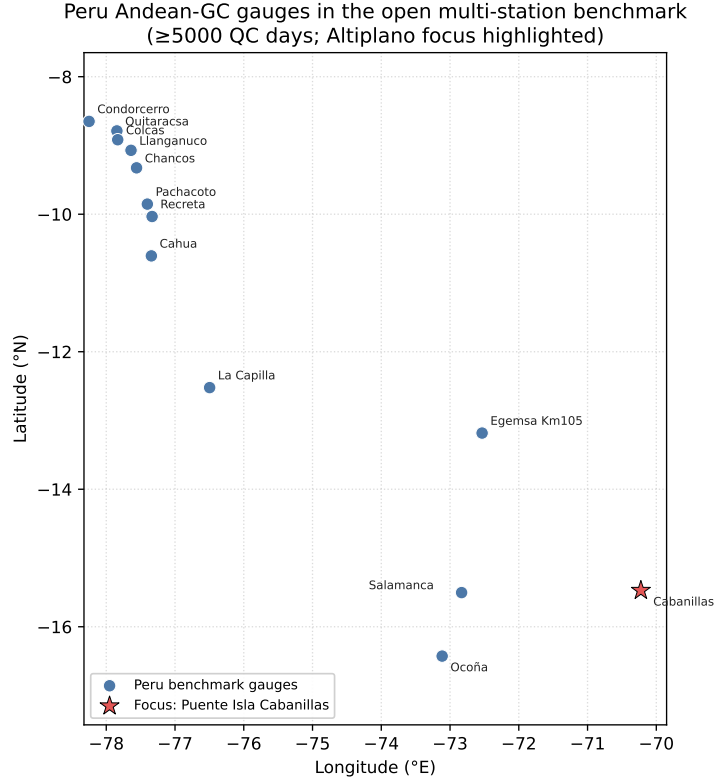


Fig. 1 Locations of the 13 Peru Andean-GC gauges with ≥ 5000 QC days used in the open benchmark. The Altiplano focus station Puente Isla Cabanillas (P00000055) is highlighted.

3 Data and Study Area

We extract daily quality-controlled discharge for all Peru SENAMHI gauges in Andean-GC [1] with $\text{days_w_data_qc} \geq 5000$ ($n = 13$), released on Zenodo under CC BY 4.0. The Altiplano focus station is P00000055 (Puente Isla Cabanillas; approximately 15.47° S, 70.22° W; Coata–Cabanillas / Titicaca system). Southern Peru gauges in the set also include Salamanca and Ocoña. Daily precipitation at each gauge latitude/longitude is obtained from the Open-Meteo historical API (ERA5-Land) [7, 8]. For Cabanillas, the modelling window after feature alignment spans 1994-08-15 to 2024-12-30. Figure 1 maps the 13 gauges; Fig. 2 shows the observed hydrograph with chronological train, validation and test segments.

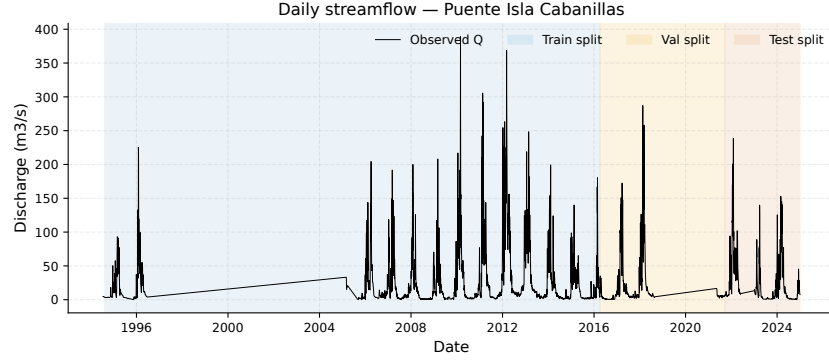


Fig. 2 Daily discharge at Puente Isla Cabanillas with chronological train, validation and test periods.

4 Methods

4.1 Features

For lead times $h \in \{1, 7\}$ days we predict Q_{t+h} from 21 predictors: Fourier seasonality (sin/cos of day-of-year), discharge lags $\{1, 2, 3, 7, 14, 30\}$, rolling means and standard deviations over $\{7, 14, 30\}$ days, and precipitation lags/rolls $\{0, 1, 2, 3, 7\}$ and $\{7, 30\}$.

4.2 Models and baselines

Baselines: persistence ($Q_{t+h} = Q_t$) and seasonal climatology (mean discharge by day-of-year on the training set). Learners: Ridge regression, Random Forest, Gradient Boosting, XGBoost and a small multilayer perceptron. All fits run on a commodity laptop CPU (full multi-station run ≈ 126 s).

4.3 Evaluation

We use a chronological 70/15/15 split (train/validation/test) with no shuffling on every station. Test metrics are RMSE, MAE, NSE, KGE [3] and skill versus persistence,

$$\text{skill} = 1 - \frac{\text{RMSE}_{\text{model}}}{\text{RMSE}_{\text{persistence}}}. \quad (1)$$

Table 2 Test-set metrics at Puente Isla Cabanillas (chronological split). Skill is relative to persistence.

Model	h	RMSE	MAE	NSE	KGE	Skill
Persistence	1	11.07	4.69	0.902	0.951	0.000
Climatology	1	28.47	15.98	0.353	0.506	-1.571
Ridge	1	13.84	8.08	0.847	0.895	-0.250
RandomForest	1	13.32	6.51	0.858	0.905	-0.203
GradientBoosting	1	12.82	6.63	0.869	0.896	-0.158
MLP	1	12.69	6.55	0.871	0.912	-0.146
XGBoost	1	12.95	6.50	0.866	0.901	-0.169
Persistence	7	25.96	12.57	0.462	0.731	0.000
Climatology	7	28.50	15.93	0.352	0.509	-0.098
Ridge	7	23.74	14.28	0.550	0.674	0.086
RandomForest	7	27.63	14.60	0.391	0.626	-0.064
GradientBoosting	7	25.64	13.95	0.475	0.639	0.012
MLP	7	24.45	13.40	0.523	0.679	0.058
XGBoost	7	27.16	14.74	0.412	0.626	-0.046

Positive skill means improvement over persistence. On the focus gauge we additionally run an expanding-window walk-forward protocol (three folds) for Ridge versus persistence. We also compute SSI-30 as the month-wise z -score of the 30-day rolling mean discharge (parametric approximation; Fig. 7). This is not the full Vicente-Serrano distributional transform and should not be treated as an operational SSI product [10].

5 Results

5.1 Focus station (Cabanillas)

Table 2 summarises the held-out chronological test period at Puente Isla Cabanillas (2021-09-24–2024-12-30; $n_{\text{test}} = 982$). At $h = 1$, persistence achieves the best NSE (0.902; KGE = 0.951); the best ML model is MLP (NSE = 0.871, KGE = 0.912) but does not beat persistence on RMSE. At $h = 7$, Ridge attains NSE = 0.550 (KGE = 0.674) versus persistence NSE = 0.462 (skill = 0.086), with MLP close behind. Expanding-window walk-forward on Cabanillas confirms positive Ridge skill at $h = 7$ in all three folds (mean skill = 0.100).

5.2 Multi-station Peru benchmark

Across all 13 gauges at $h = 7$, the best ML model per station yields positive skill versus persistence in 13/13 cases (median skill = 0.102). Ridge alone is skill-

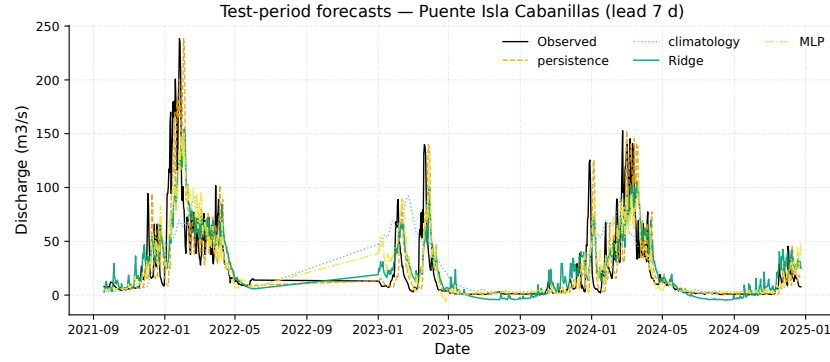


Fig. 3 Observed versus forecast discharge on the Cabanillas test set at a 7-day lead.

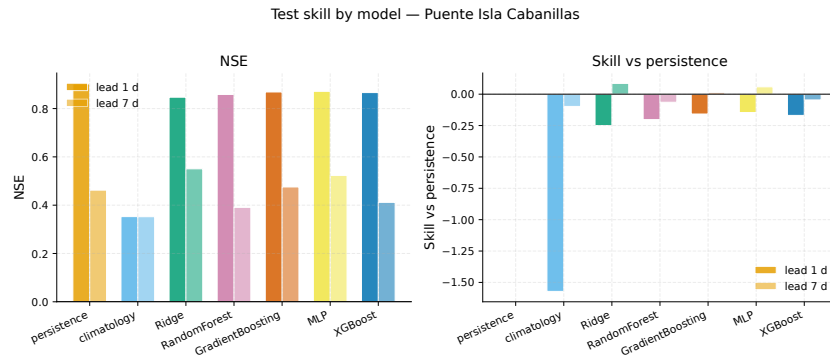


Fig. 4 Skill versus persistence by model and lead time at Puente Isla Cabanillas. Positive values favour the learner.

positive at every station (mean NSE = 0.639 vs. persistence mean NSE = 0.569; mean skill = 0.078; mean KGE = 0.714). Figures 5–6 summarise cross-station skill and NSE.

6 Discussion

The one-day result is a feature, not a failure: when autocorrelation is strong, claiming “ML superiority” without a persistence baseline would be misleading. The seven-day gain of Ridge and MLP is modest but reproducible across the Peru gauge set and obtained with open data and minutes of CPU time. Walk-forward folds on Cabanillas support the chronological $h = 7$ skill signal.

Limitations include: (i) reanalysis precipitation rather than local rain gauges; (ii) no routing or process constraints; (iii) SSI-30 as a z -score approximation rather

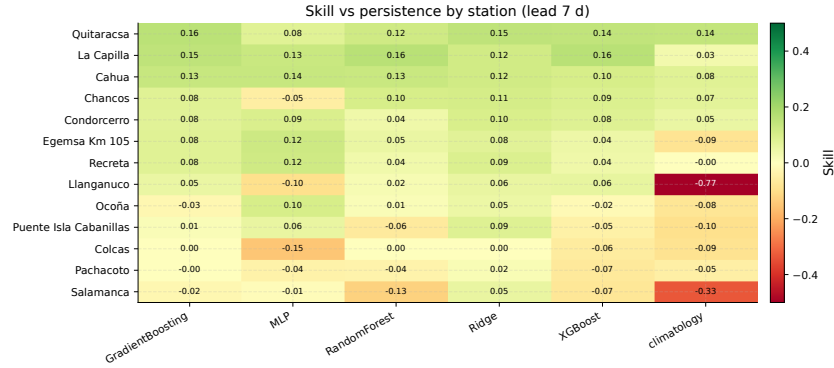


Fig. 5 Multi-station skill versus persistence at a 7-day lead (Peru Andean-GC gauges with ≥ 5000 QC days).

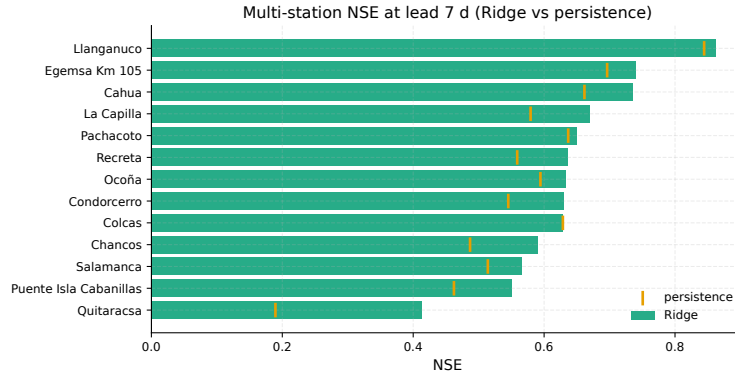


Fig. 6 Multi-station NSE at a 7-day lead for Ridge versus persistence markers.

than a full distributional SSI [10]; (iv) heterogeneous record eras among Cordillera Blanca historical gauges versus modern Altiplano records; (v) no formal uncertainty quantification or operational data-assimilation loop. Multi-basin transfer learning, local SENAMHI precipitation and full SSI implementations are natural extensions. This work does not replace national hydrological services or institutional warning systems.

7 Conclusion

We present an open multi-station Peru streamflow forecasting benchmark with Altiplano focus, honest baselines, NSE/KGE reporting, SSI-30 drought indication and Springer-ready evidence. At a one-day lead, persistence dominates; at seven

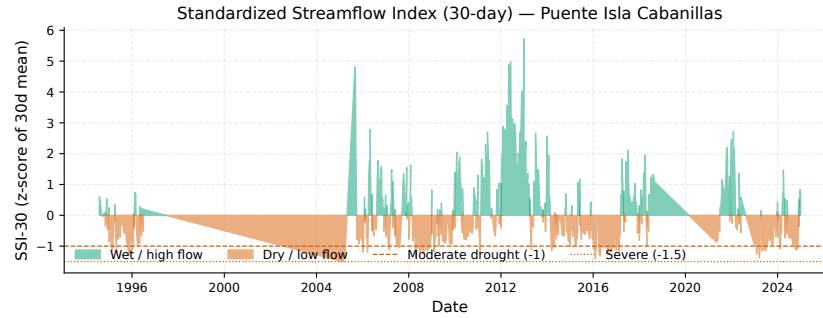


Fig. 7 SSI-30 at Puente Isla Cabanillas: month-wise z-score of the 30-day rolling mean (parametric approximation).

days, lightweight linear and neural models provide positive skill on all 13 long gauges tested. Code, metrics and figures are hosted in a public GitHub repository: `\githubUrl{}`.

Data availability

Daily QC discharge is from Andean-GC v1.1 on Zenodo (CC BY 4.0; doi:<https://doi.org/10.5281/zenodo.18035801>). Daily precipitation was retrieved from the Open-Meteo Historical Weather API (ERA5-Land). Station extracts, precip caches and processed tables are documented in the project inventory file `INVENTARIO.md` under `02-datos/fuentes/`.

Code availability

Analysis code (feature construction, models, metrics, figures) is provided under an MIT license in the public GitHub repository: `\githubUrl{}`. Reproduce with `python 03-codigo/run_experiment.py` after installing `requirements.txt`.

Author contributions

R.A.V.-S.: conceptualization, data curation, methodology, software, formal analysis, visualization, writing—original draft. D.M.Y.-Y.: methodology, validation, writing—review and editing. Both authors approved the final manuscript.

Competing interests

The authors declare no competing interests.

Ethics and limitations

No human subjects or personal data were used. Streamflow and reanalysis fields are public open data. Forecasts are research benchmarks only and must not be used as sole input for flood or drought emergency decisions without institutional validation.

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